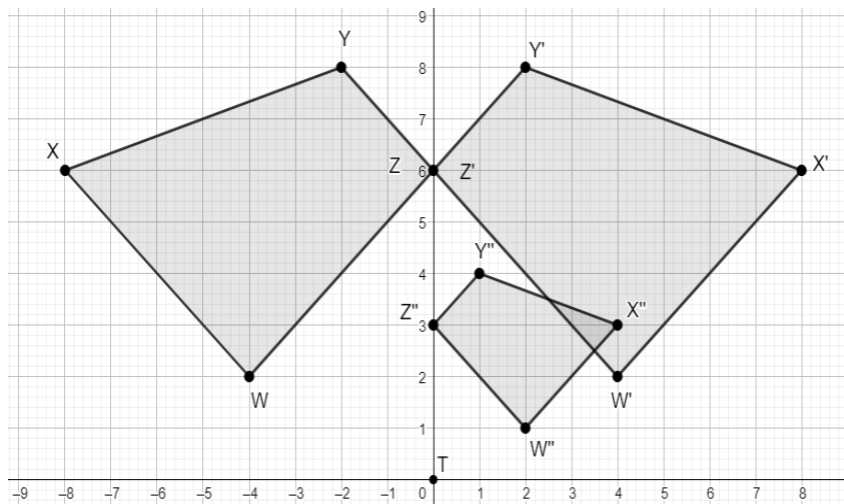


Geometry
Unit 6
Lesson 2 Practice

Name _____
Date _____
Period _____

A sequence of transformations of $WXYZ$ is shown on the coordinate grid. $WXYZ$ is reflected then dilated, where $W'X'Y'Z'$ is the reflection and $W''X''Y''Z''$ is the dilation of $W'X'Y'Z'$.



Complete the descriptions:

- $WXYZ$ was _____ to create $W'X'Y'Z'$.
- $W'X'Y'Z'$ was dilated using a scale factor of _____ centered at _____ to create $W''X''Y''Z''$.
- Complete the table:

Transformation	Algebraic Description
Reflection	$(x, y) \rightarrow$
Dilation	$(x, y) \rightarrow$

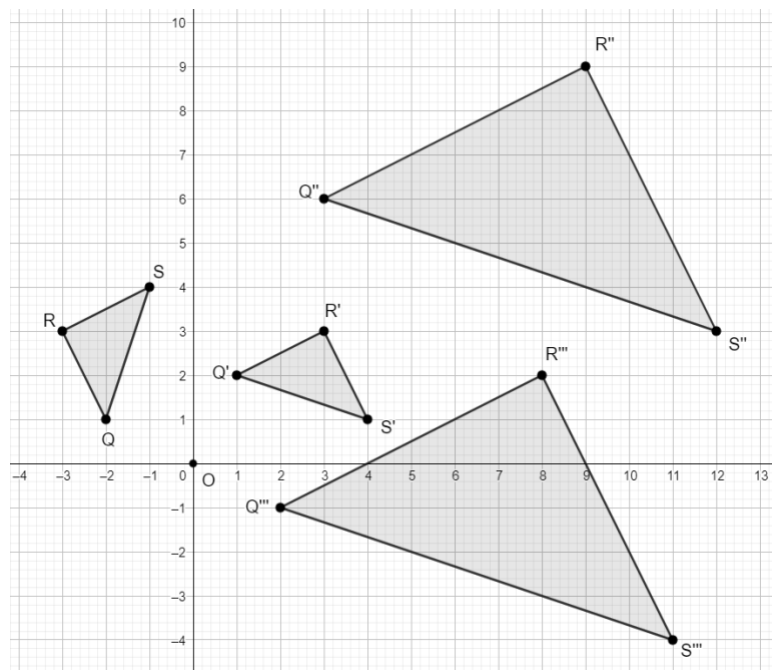
- Determine the coordinates if $WXYZ$ is first dilated then reflected:

Preimage $WXYZ$		Dilation $W'X'Y'Z'$		Reflection $W''X''Y''Z''$	
W	$(-4, 2)$	W'		W''	
X	$(-8, 6)$	X'		X''	
Y	$(-2, 8)$	Y'		Y''	
Z	$(0, 6)$	Z'		Z''	

- Based on the results above, the sequence of transformations commutative.

☐ is
☐ is not

$QRS, Q'R'S', Q''R''S'',$ and $Q'''R'''S'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- QRS is first

- ☐ translated
 - ☐ reflected
 - ☐ rotated
 - ☐ dilated

 to $Q'R'S'$, then

- ☐ translated
 - ☐ reflected
 - ☐ rotated
 - ☐ dilated

 to $Q''R''S''$,
- and finally

- ☐ translated
 - ☐ reflected
 - ☐ rotated
 - ☐ dilated

 to $Q'''R'''S'''$.
7. Describe the transformation for $QRS \rightarrow Q'R'S'$.
8. Describe the transformation for $Q'R'S' \rightarrow Q''R''S''$.
9. Describe the transformation for $Q''R''S'' \rightarrow Q'''R'''S'''$.

10. Write the algebraic description for the sequence of transformations from 7-9:

Transformation	Algebraic Description
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$